

## Chapter 03 Homework — Rings

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September 7, 2026

**Problem 03.1.** Suppose that  $S$  is a set and  $R$  is a ring. Let  $R^S$  denote the set of all mappings from  $S$  to  $R$ . Show that  $R^S$  is a ring, under the operations defined by

$$(f + g)(s) = f(s) + g(s), \quad (fg)(s) = f(s)g(s).$$

Show that if  $S$  has more than one element then there exist non-zero elements  $f$  and  $g$  in  $R^S$  for which  $fg = 0$ .

**Solution.** Write  $f_1, f_0 \in R^S$  for the functions

$$f_1(s) = 1_R, \quad f_0(s) = 0_R,$$

and clearly, by the specified function properties, we have all ring properties.

Prove  $(f + g)h = fh + gh$ . Let  $s \in S$ . Show  $((f + g)h)(s) = (fh)(s) + (gh)(s)$ :

$$((f + g)h)(s) = (f(s) + g(s))h(s) = f(s)h(s) + g(s)h(s) = (fh)(s) + (gh)(s).$$

If  $|S| > 1$ , there exist  $s_1, s_2 \in S$  where  $s_1 \neq s_2$ . Let

$$f(s) = \begin{cases} 1_R & s = s_1 \\ 0_R & \text{else} \end{cases} \quad g(s) = \begin{cases} 1_R & s = s_2 \\ 0_R & \text{else} \end{cases}$$

Then  $f(s)g(s) = 0$ , so they are zero divisors.

**Problem 03.2.** Suppose that  $R$  is an integral domain, with field of fractions  $F$ . Show that the field of fractions of  $R[x_1, \dots, x_n]$  can be identified naturally with  $F(x_1, \dots, x_n)$ .

**Problem 03.3.** *[TODO: statement not yet transcribed]*

**Problem 03.8.** Suppose that  $K$  is a field. If  $f = a_0 + a_1x + \dots + a_nx^n \in K[x]$  and  $k \in K$ , let  $(\Phi(f))(k) = a_0 + a_1k + \dots + a_nk^n$ . Show that  $\Phi$  is a ring homomorphism from  $K[x]$  to  $K^K$ . Show that if  $K$  is finite then  $\Phi$  is an epimorphism, but not a monomorphism. What happens if  $K$  is infinite?

**Solution.** Let  $f = \sum_i a_i x^i$  and  $g = \sum_i b_i x^i$  in  $K[x]$ .

Additive half. Let  $k \in K$ . Then

$$\begin{aligned} (\Phi(f + g))(k) &= \left( \Phi\left(\sum_i (a_i + b_i)x^i\right) \right)(k) = \sum_i (a_i + b_i)k^i = \sum_i a_i k^i + \sum_i b_i k^i \\ &= \left( \Phi\left(\sum_i a_i x^i\right) \right)(k) + \left( \Phi\left(\sum_i b_i x^i\right) \right)(k) = (\Phi(f))(k) + (\Phi(g))(k). \end{aligned}$$

As  $k \in K$  was arbitrary,  $\Phi(f + g) = \Phi(f) + \Phi(g)$  in  $K^K$ .

*Multiplicative half.* Let  $k \in K$ . Then

$$\begin{aligned} (\Phi(fg))(k) &= \left( \Phi \left( \sum_m \left( \sum_{i+j=m} a_i b_j \right) x^m \right) \right)(k) = \sum_m \left( \sum_{i+j=m} a_i b_j \right) k^m \\ &= \left( \sum_i a_i k^i \right) \left( \sum_j b_j k^j \right) = \left( \Phi \left( \sum_i a_i x^i \right) \right)(k) \cdot \left( \Phi \left( \sum_j b_j x^j \right) \right)(k) = (\Phi(f))(k) (\Phi(g))(k). \end{aligned}$$

$\Phi$  is an epimorphism when  $K$  is finite. Let  $K$  be a finite field and let  $h \in K^K$  be arbitrary. We find  $f \in K[x]$  with  $(\Phi(f))(k) = h(k)$ . Take

$$f(x) = \sum_k h(k) \delta_k(x).$$

Let  $k_1 \in K$ . Then

$$(\Phi(f))(k_1) = \left( \Phi \left( \sum_k h(k) \delta_k(x) \right) \right)(k_1) = \sum_k h(k) \delta_k(k_1) = h(k_1) \delta_{k_1}(k_1) = h(k_1) \cdot 1 = h(k_1).$$

So  $\Phi$  is an epimorphism.

$\Phi$  is not a monomorphism. Take  $K = \mathbb{F}_2$ ,  $f = x^2 + x$  and  $g = 0$ . Then  $f \neq g$  in  $\mathbb{F}_2[x]$ , but

$$(\Phi(f))(0) = 0 = (\Phi(g))(0), \quad (\Phi(f))(1) = 1 + 1 = 0 = (\Phi(g))(1),$$

so  $\Phi(f) = \Phi(g)$  in  $\mathbb{F}_2^{\mathbb{F}_2}$  and  $\ker \Phi \neq \{0\}$ .

For an arbitrary finite field  $K$ , take

$$f(x) = \prod_{k \in K} \delta_k(x),$$

a non-zero element of  $\ker \Phi$ .

**Problem 03.10.** What are the units in  $R[x]$ , where  $R$  is an integral domain? What are the units in  $\mathbb{Z}_4[x]$ ?

**Solution.** The units in  $R[x]$ , for  $R$  an integral domain, are

$$\{a \in R \subseteq R[x] : a^{-1} \text{ exists}\},$$

the existence of  $a^{-1}$  being what is not guaranteed in an integral domain.

They must be constants. For in an integral domain  $R$ , with  $f, g \in R[x]$ ,

$$\deg(fg) = \deg(f) + \deg(g),$$

since no non-zero coefficients multiply to zero. So  $\deg(f) = \deg(g) = 0$ .

In  $\mathbb{Z}_4[x]$  the units are 1 and 3, since  $3 \cdot 3 = 9 \equiv 1$ ; and also, taking

$$f(x) = 2x + 3, \quad g(x) = 2x + 3,$$

we have  $fg = 4x^2 + 12x + 9 = 1$ .

So in  $\mathbb{Z}_4[x]$  the units are those  $f$  whose constant term is a unit and all of whose other coefficients are zero divisors: constant term in  $\{1, 3\}$ , every other coefficient in  $\{0, 2\}$ .

**Problem 03.11.** Show that an element  $a$  of an integral domain  $R$  is prime if and only if  $R/(a)$  is an integral domain.

**Solution.** ( $\Rightarrow$ ) Suppose  $a \in R$  is prime. Show  $R/(a)$  is an integral domain.

Consider

$$(r_1 + (a))(r_2 + (a)) = r_1r_2 + (a).$$

Take the factors non-zero, so  $a \nmid r_1$  and  $a \nmid r_2$  — else we would have zero divisors. Then  $r_1r_2 + (a)$  is non-zero, as  $a \nmid r_1r_2$  since  $a$  is prime. So  $R/(a)$  has no zero divisors.

Further,  $1 + (a) \neq 0 + (a)$ , because  $a \nmid 1$  ( $a$  is not a unit).

( $\Leftarrow$ ) Suppose  $R/(a)$  is an integral domain. Show  $a \in R$  is prime.

Suppose  $a$  is not prime. So there exist  $b, c \in R$  such that  $a \mid bc$ , but  $a \nmid b$  and  $a \nmid c$ . Note  $b + (a)$  and  $c + (a)$  lie in  $R/(a)$ , both non-zero since  $a \nmid b$ ,  $a \nmid c$ . And

$$(b + (a))(c + (a)) = bc + (a) = (a)$$

since  $a \mid bc$ . So  $R/(a)$  has zero divisors and is not an integral domain.

Further,  $a$  is not a unit. Suppose  $a \in R$  is a unit; show  $R/(a)$  is not an integral domain. Note  $a^{-1} \in R$ , so  $aa^{-1} = e \in (a)$ . Let  $r \in R$ . Note  $r \in (a)$  if there exists  $x \in R$  such that  $xa = r$ ; take  $x = ra^{-1}$ , which exists since  $a$  is a unit in  $R$ . So  $(a) = R$ , and

$$R/(a) = R/R = \{R\},$$

a single-element set, which cannot be an integral domain since it is a ring with  $e = 0$ .

**Problem 03.12.** Let  $R = \mathbb{Z} + i\sqrt{5}\mathbb{Z}$ .

1. Show that the units are 1 and  $-1$ .
2. Let  $\varphi(m + i\sqrt{5}n) = m^2 + 5n^2$ . Then  $\varphi(0) = 0$ ,  $\varphi(1) = \varphi(-1) = 1$  and otherwise  $\varphi(\alpha) > 3$ . Use this to show that  $2 + i\sqrt{5}$  and  $2 - i\sqrt{5}$  are irreducible in  $R$ .
3. Show that  $2 + i\sqrt{5}$  is not a prime;  $R$  is not a unique factorization domain.

**Solution.** (a) Both 1 and  $-1$  are units, each being its own inverse.

Conversely, suppose  $\alpha \in R$  is a unit, so  $\alpha\beta = 1$  for some  $\beta \in R$ . Then

$$\varphi(\alpha\beta) = \varphi(1) = 1,$$

and so  $\varphi(\alpha)\varphi(\beta) = 1$ . Since  $\varphi : R \rightarrow \mathbb{N}$ , this gives  $\varphi(\alpha) = 1$  and  $\varphi(\beta) = \varphi(\alpha)$ . Through the definition of  $\varphi$ , this forces  $m = \pm 1$ ,  $n = 0$ ; that is,  $\alpha = \pm 1$ .

**Problem 03.14.** *[TODO: statement not yet transcribed]*

**Problem 03.15.** *[TODO: statement not yet transcribed]*

**Problem 03.17.** *[TODO: statement not yet transcribed]*

**Problem 03.25.** *[TODO: statement not yet transcribed]*

**Problem 03.26.** *[TODO: statement not yet transcribed]*